



The lambda number of the power graph of a finite p -group

Siddhartha Sarkar¹ · Mayank Mishra²

Received: 11 June 2021 / Accepted: 29 July 2022 / Published online: 19 September 2022

© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2022

Abstract

An $L(2, 1)$ -labelling of a finite graph Γ is a function that assigns integer values to the vertices $V(\Gamma)$ of Γ (colouring of $V(\Gamma)$ by $\mathbb{Z}$) so that the absolute difference of two such values is at least 2 for adjacent vertices and is at least 1 for vertices, which are precisely distance 2 apart. The lambda number $\lambda(\Gamma)$ of Γ measures the least number of integers needed for such a labelling (colouring). A power graph Γ_G of a finite group G is a graph with vertex set as the elements of G and two vertices are joined by an edge if and only if one of them is a positive integer power of the other. It is known that $\lambda(\Gamma_G) \geq |G|$ for any finite group. In this paper, we show that if G is a finite group of a prime power order, then $\lambda(\Gamma_G) = |G|$ if and only if G is neither cyclic nor a generalized quaternion 2-group. This settles a partial classification of finite groups achieving the lower bound of lambda number.

Keywords Power graph · $L(2, 1)$ -labelling · λ -number · Finite p -group

Mathematics Subject Classification 05C25 · 05C78 · 20D15

1 Introduction

All graphs considered in this article are assumed to be finite, undirected and simple. Let Γ be a graph with vertex set $V(\Gamma)$ and edge set $E(\Gamma)$. For any $a, b \in V(\Gamma)$, the distance $d(a, b)$ is defined to be the length of the shortest path joining the vertices a and b in Γ . For any $j, k \in \mathbb{N}_0 := \mathbb{N} \cup \{0\}$, an $L(j, k)$ -labelling is defined to be a function $f : V(\Gamma) \rightarrow \mathbb{Z}$ that satisfies:

- (i) $|f(a) - f(b)| \geq j$ whenever $d(a, b) = 1$,
- (ii) $|f(a) - f(b)| \geq k$ whenever $d(a, b) = 2$.

✉ Siddhartha Sarkar
sidhu@iiserb.ac.in

¹ Department of Mathematics, Indian Institute of Science Education and Research Bhopal, Bhopal Bypass Road, Bhauri, Bhopal, Madhya Pradesh 462 066, India

² Kalyanpur, Kanpur Nagar, India

Therefore, an $L(1, 0)$ -labelling is simply a colouring of the graph Γ by non-negative integers. For an $L(j, k)$ -labelling f on Γ , the span of f is defined by

$$\text{span}(f) := \max_{x \in V(\Gamma)} f(x) - \min_{x \in V(\Gamma)} f(x).$$

The $L(j, k)$ -labelling number $\lambda_{j,k}$ is defined as:

$$\lambda_{j,k}(\Gamma) := \min \left\{ \text{span}(f) : f \text{ is an } L(j, k)\text{-labelling on } \Gamma \right\}.$$

The *lambda number* of the graph Γ is defined by $\lambda(\Gamma) := \lambda_{2,1}(\Gamma)$.

The $L(j, k)$ -labelling of graphs was motivated by the study of the radio channel assignment problem of transmitters through radio or television channels so that the recipients are needed to be separated by certain constraints due to interference (cf. [8, 18]). One can see [19] for a complete survey related to the problem and the references therein.

The concepts of power graph and undirected power graph were first introduced by Kelarev and Quinn [10] and Chakrabarty et al. [6], respectively. Now a power graph means an undirected power graph. In recent years, the study of power graphs has been growing. The structural behaviour of power graphs of finite groups is studied in [4, 5]. In fact, the strong metric dimension of the power graph of a finite group was characterized in [13] (see also [16]) and the properties of power graphs of (non)orientable genus two were studied in [15]. For more information about the recent progress on the study of power graphs, see [1, 12].

This article is related to the study of lambda number of the power graph Γ_G of a finite group G . This graph is defined by taking the elements of G as its vertex set, and two distinct vertices $a, b \in G$ are connected by an edge if and only if either $a = b^n$ or $b = a^m$ for some $m, n \in \mathbb{N}$. The notion of (directed) power graphs were introduced in [6] while studying semigroups. Later on, it was shown in a sequel of articles [4, 5] that the power graphs of two finite groups are isomorphic if and only if they contain the same number of elements of any given order. In this article, we will be focussing on finite groups, and consequently, Γ_G is always undirected.

In [14], it is shown that $\lambda(\Gamma_G) \geq |G|$ for any finite group G and the following question is asked:

Question. Classify all finite groups G which satisfying $\lambda(\Gamma_G) = |G|$.

Let $p \in \mathbb{N}$ be a prime. A finite group G is called a p -group if $|G| = p^n$ for some $n \in \mathbb{N}$. For such a group G , its exponent is the maximum of the orders of elements of G . The complete classification of these groups is not known till date. The main problem is the set of parameters those describe the presentations (or even from the viewpoint of representations given over a number field) gets too complicated very quickly as $n = \log_p |G|$ become large.

The main theorem of this paper is as follows:

Main Theorem. (see Corollary 2) Let G be a finite p -group. Then $\lambda(\Gamma_G) = |G|$ if and only if G is neither cyclic, nor a generalized quaternion 2-group.

Consequently, for a finite p -group G of exponent p^e we have

$$\lambda(\Gamma_G) = \begin{cases} 2(p^e - 1) & \text{if } G \text{ is cyclic} \\ |G| + 1 & \text{if } p = 2, G \text{ is generalized quaternion} \\ |G| & \text{otherwise} \end{cases}$$

We discussed some basic group theoretic preliminaries in Sect. 2. The proof of the main theorem is given in Sect. 3.

2 Preliminaries

Definition 1 Let G be a finite group and n be a positive integer so that G contains an element of order n . We define the set $\Lambda(n; G)$ (or simply by $\Lambda(n)$) as

$$\Lambda(n; G) := \{g \in G : |g| = n\}.$$

where $|g|$ denote the order of g .

Given any two elements $g_1, g_2 \in G$, we define $g_1 \sim g_2$ if the cyclic subgroups of G generated by g_1 and g_2 coincide. This is an equivalence relation on G . An equivalence class containing $g \in G$ is denoted by $[g]$, and we call it a **cyclic class** of G . Then, we have

$$\Lambda(n; G) = \bigcup_{|g|=n} [g]$$

where the union runs over a set of class representatives $g \in G$ with $|g| = n$. In particular, if n is a power of a prime, then the elements of the class $[g]$ induce a complete subgraph of the power graph Γ_G .

The class $\{1_G\}$ is called the **trivial cyclic class**, where 1_G denote the identity element of the group G . For any non-trivial cyclic class $[g]$ with $|g| = n$, the class $[g]$ contain $\phi(n)$ number of elements, where ϕ denotes the Euler's phi function. For any $n \in \mathbb{N}$, we define the **n -th cyclic class number** $m(n)$ as the number of cyclic classes that contain the elements of order n , i.e.

$$m(n) := \frac{|\Lambda(n; G)|}{\phi(n)}.$$

Note 1 The n -th cyclic class number $m(n)$ of a group G is equal to the number of cyclic subgroups of G of order n . For example, if G is a cyclic group, then $m(n) = 1$ for every n divides $|G|$.

Let G be a group and $H, K \leq G$ be subgroups. The subgroup $[H, K] \leq G$ is defined as the subgroup generated by all elements of the form $[h, k] := h^{-1}k^{-1}hk$ ($h \in$

$H, k \in K$). The lower central series of subgroups of G is the descending sequence

$$G \geq \gamma_2(G) \geq \gamma_3(G) \geq \dots \geq \gamma_i(G) \geq \gamma_{i+1}(G) \geq \dots$$

of normal subgroups of G given by $\gamma_2(G) := [G, G]$ and $\gamma_{i+1}(G) := [\gamma_i(G), G]$ for every $i \geq 2$. If G is a finite p -group, then this descending sequence contains only finitely many non-trivial terms (in general such groups are called *nilpotent*).

A finite p -group of order p^n is said to be of *maximal class* if $\gamma_{n-1}(G) \neq \{1_G\}$ and $\gamma_n(G) = \{1_G\}$. In this case, $G/\gamma_2(G) \cong C_p \times C_p$ and $\gamma_i(G)/\gamma_{i+1}(G) \cong C_p$ for all $2 \leq i \leq n-1$.

The following result says a lot more about the class numbers of a finite p -group G . For a complete proof, see [2, Theorems 1.10, 1.17].

Theorem 1 (see [3, 9, 11, 17]) *Let G be a finite p -group of exponent p^e . Assume that G is not cyclic for an odd prime p , and for $p = 2$, it is neither cyclic nor of maximal class. Then,*

- (i) $m(p) \equiv 1 + p \pmod{p^2}$.
- (ii) $p \mid m(p^i)$ for every $2 \leq i \leq e$.

Corollary 1 *Let G be a finite p -group of exponent p^e . Then, $m(p^i) = 1$ for some $1 \leq i \leq e$ if and only if one of the following occurs:*

- (i) $G \cong C_{p^e}$ and $m(p^j) = 1$ for all $1 \leq j \leq e$, or
- (ii) $p = 2$ and G is isomorphic to one of the following 2-groups:
 - (a) *dihedral 2-group*

$$\mathbb{D}_{2^{e+1}} = \langle x, y : x^{2^e} = 1, y^2 = 1, y^{-1}xy = x^{-1} \rangle, \quad (e \geq 2)$$

where $m(2) = 1 + 2^e$ and $m(2^j) = 1$ ($2 \leq j \leq e$).

- (b) *generalized quaternion 2-group*

$$\mathbb{Q}_{2^{e+1}} = \langle x, y : x^{2^e} = 1, x^{2^{e-1}} = y^2, y^{-1}xy = x^{-1} \rangle, \quad (e \geq 2)$$

where $m(4) = 1 + 2^{e-1}$ and $m(2^j) = 1$ for all $2 \leq j \leq e$ and $j \neq 2$.

- (c) *semi-dihedral 2-group*

$$\mathbb{SD}_{2^{e+1}} = \langle x, y : x^{2^e} = 1, y^2 = 1, y^{-1}xy = x^{-1+2^{e-1}} \rangle, \quad (e \geq 3)$$

where $m(2) = 1 + 2^{e-1}$, $m(4) = 1 + 2^{e-2}$ and $m(2^j) = 1$ for all $3 \leq j \leq e$.

Proof From Theorem 1, it follows that if the hypothesis holds then G is either cyclic, or else G is a 2-group of maximal class. From [7, Theorem 5.1], G must be isomorphic to one of the groups given in (ii)(a)–(c).

To establish that the condition is necessary and sufficient, we only need to verify the values of $m(2^j)$ ($1 \leq j \leq e$) as given. Since a cyclic group has a unique subgroup of

order 2^j for every $1 \leq j \leq e$, the statement (i) follows. Notice that each group given in (ii) has a cyclic maximal subgroup $\langle x \rangle$ of order 2^e .

(ii)(a) For $G = \mathbb{D}_{2^{e+1}}$, every element outside the maximal cyclic subgroup $\langle x \rangle \leq G$ has order 2. Counting the element $x^{2^{e-1}} \in \langle x \rangle$ of order 2, we have $m(2) = 1 + 2^e$. For $2 \leq j \leq e$, an element of order 2^j is a generator of the cyclic subgroup $\langle x^{2^{e-j}} \rangle$ and hence $m(2^j) = 1$.

(ii)(b) The group $G = \mathbb{Q}_{2^{e+1}}$ has a unique element of order 2, namely $x^{2^{e-1}}$. Since every element outside $\langle x \rangle$ has order 4 and $\phi(4) = 2$, with a similar argument as above, the statement follows.

(ii)(c) For $G = \mathbb{SD}_{2^{e+1}}$, every element outside $\langle x \rangle$ can be uniquely written as $x^k y$ ($0 \leq k \leq 2^e - 1$). From the given relations, it is straightforward to verify

$$|x^k y| = \begin{cases} 2 & \text{if } k \text{ is even} \\ 4 & \text{if } k \text{ is odd} \end{cases}.$$

Counting $x^{2^{e-1}}$, then we have exactly $1 + 2^{e-1}$ of elements of order 2. Similarly, counting the two elements of order 4 from $\langle x^{2^{e-2}} \rangle$, we have $2 + 2^{e-1}$ elements of order 4 and these form a disjoint union of classes of size $\phi(4) = 2$. Now all elements of order ≥ 8 must belong to $\langle x \rangle$. $\square$

We now look at a result from [14] and obtain some insights that are going to make the proofs of the next section shorter and easier to understand.

Definition 2 Let Γ be a graph. The **complement graph** Γ^c is defined as the graph with the same vertex set $V(\Gamma)$ and two vertices $a, b \in \Gamma^c$ are joined by an edge if and only if they are not adjacent in Γ .

Theorem 2 ([14, Theorem 3.1]) *Let G be finite group. Then $\lambda(\Gamma_G) \geq |G|$, with equality if and only if the complement graph $(\Gamma_G \setminus \{1_G\})^c$ contains a Hamiltonian path.*

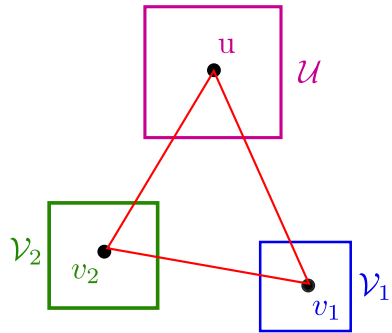
Let G be a finite group G with $\lambda(\Gamma_G) = |G|$. We set $N := |G| - 2$. If $(x_0, x_1, \dots, x_N)$ is a Hamiltonian path in $(\Gamma_G \setminus \{1_G\})^c$, then the function $f : G \rightarrow \mathbb{Z}$ given by $f(1_G) := -2$ and $f(x_i) := i$ ($0 \leq i \leq N$) defines an $L(2, 1)$ -labelling of Γ_G that achieves the lower bound $\text{span}(f) = |G|$.

On the other hand, if $f : G \rightarrow \mathbb{Z}$ is an $L(2, 1)$ -labelling of G with $\text{span}(f) = |G|$, then the same holds for a translation map $f + \alpha$ for any $\alpha \in \mathbb{Z}$. Now by definition f is an injective map, and from Pigeonhole principle, it follows that the image $\text{Im}(f)$ of the function f contains precisely $|G| + 1$ integers with $f(1_G)$ is either the maximum or minimum of $\text{Im}(f)$. In case $f(1_G)$ is a maximum of $\text{Im}(f)$, we can define an $L(2, 1)$ -labelling f_1 by

$$f_1(x) = \begin{cases} \min \text{Im}(f) - 2 & \text{if } x = 1_G \\ f(x) & \text{if } x \neq 1_G. \end{cases}$$

Further by applying a translation, we can assume that $f_1(1_G) = -2$ and $\text{Im}(f) = \{-2, 0, 1, \dots, N\}$. Then, the ordered sequence $(f_1^{-1}(0), f_1^{-1}(1), \dots, f_1^{-1}(N))$ of vertices in $G \setminus \{1_G\}$ is a Hamiltonian path in $(\Gamma_G \setminus \{1_G\})^c$.

Fig. 1 Lower hook diagram



3 Proof of the main theorem

We need some preparation before we discuss the proof of the main result.

Definition 3 Let G be a non-trivial finite group. Two distinct cyclic classes $\mathcal{C}_1$ and $\mathcal{C}_2$ of G are called **adjacent** if there are elements $g_1 \in \mathcal{C}_1$ and $g_2 \in \mathcal{C}_2$ so that g_1 and g_2 are adjacent in the power graph Γ_G .

Clearly, if the cyclic classes $\mathcal{C}_1$ and $\mathcal{C}_2$ are adjacent, then any pair of elements $x_1 \in \mathcal{C}_1$ and $x_2 \in \mathcal{C}_2$ are adjacent in Γ_G . Moreover, if the cyclic classes $\mathcal{C}_1$ and $\mathcal{C}_2$ are adjacent and are represented by elements of the same order, then $\mathcal{C}_1 = \mathcal{C}_2$.

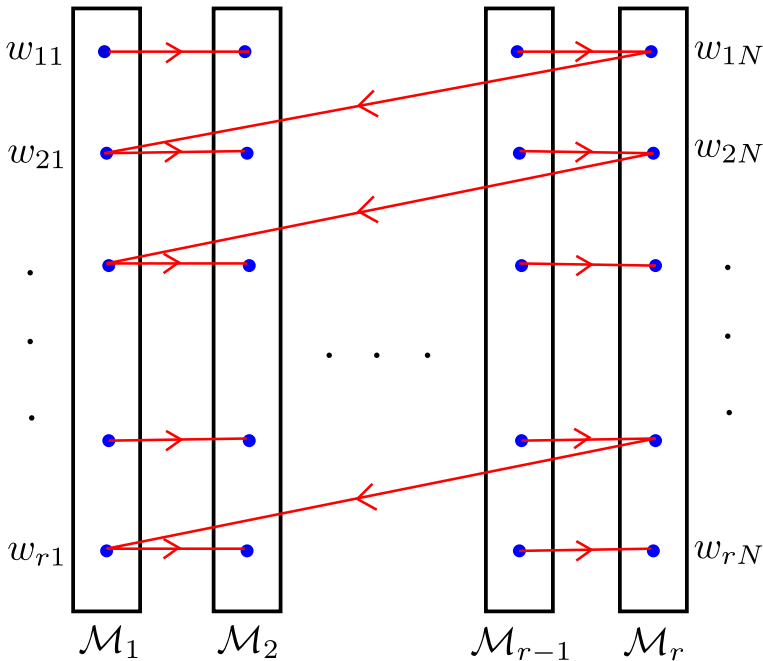
Lemma 1 (Lower hook lemma) *Let G be a finite p -group and $\mathcal{U}, \mathcal{V}_1$ and $\mathcal{V}_2$ be three classes represented by elements u, v_1 and v_2 , respectively, so that $\max\{|v_1|, |v_2|\} \leq |u|$. If $\mathcal{U}$ is adjacent to both of $\mathcal{V}_1$ and $\mathcal{V}_2$, then $\mathcal{V}_1$ is adjacent to $\mathcal{V}_2$. In particular, if $|v_1| = |v_2|$, then $\mathcal{V}_1 = \mathcal{V}_2$.*

Proof Since u is adjacent to v_1 and $|v_1| \leq |u|$, we have $\langle v_1 \rangle \subseteq \langle u \rangle$ (see Fig. 1). Similarly, $\langle v_2 \rangle \subseteq \langle u \rangle$. Without loss of generality, assume that $|v_1| \leq |v_2|$. Since $\langle u \rangle$ is a cyclic group of prime power order, it follows that $|v_1|$ divides $|v_2|$ and consequently $\langle v_1 \rangle \subseteq \langle v_2 \rangle \subseteq \langle u \rangle$. Now the conclusion follows from the fact that the elements of $\langle u \rangle$ induce a complete subgraph of Γ_G . $\square$

Note 2 The lower hook lemma 1 is not true for arbitrary finite groups. The cyclic group C_6 of order 6 has four classes $\mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3$ and $\mathcal{C}_6$ represented by elements of orders 1, 2, 3 and 6, respectively. Here $\mathcal{C}_6$ is adjacent to both of $\mathcal{C}_2$ and $\mathcal{C}_3$, but the classes $\mathcal{C}_2$ and $\mathcal{C}_3$ are not adjacent.

Definition 4 Let Γ be a graph and $\mathfrak{M} := (\mathcal{M}_1, \mathcal{M}_2, \dots, \mathcal{M}_r)$ be an ordered collection of subsets of $V(\Gamma)$ of equal size N for some $r \geq 2$. Assume that:

- (i) each $\mathcal{M}_i$ induces a complete subgraph of Γ , and
- (ii) for any $1 \leq i \neq j \leq r$ and any $a \in \mathcal{M}_i, b \in \mathcal{M}_j$, the vertices a and b are not adjacent in Γ .

Fig. 2 $\mathcal{H}(\mathfrak{M})$

We order the elements of $\mathcal{M}_j$ as follows:

$$\mathcal{M}_j = \{w_{j1}, w_{j2}, \dots, w_{jN}\}, \quad (j = 1, \dots, r).$$

We define the Hamiltonian path $\mathcal{H}(\mathfrak{M})$ on the set $\cup_{j=1}^N \mathcal{M}_j$ of vertices (see Fig. 2) as a subgraph of Γ^c by:

$$\mathcal{H}(\mathfrak{M}) := (w_{11}, w_{21}, \dots, w_{r-1,1}, w_{r1}, \dots, w_{1N}, w_{2N}, \dots, w_{r-1,N}, w_{rN})$$

We denote by $\partial_0(\mathcal{H}(\mathfrak{M})) := w_{11}$ and $\partial_1(\mathcal{H}(\mathfrak{M})) := w_{rN}$, the initial and the terminal vertices of the Hamiltonian path $\mathcal{H}(\mathfrak{M})$.

Theorem 3 Let G be a finite p -group of exponent p^e with $m(p^i) \geq 2$ for every $1 \leq i \leq e$. Then $\lambda(\Gamma_G) = |G|$.

Proof For each $1 \leq i \leq e$, we set $m_i := m(p^i)$ and let $\mathcal{C}_1^{(i)}, \mathcal{C}_2^{(i)}, \dots, \mathcal{C}_{m_i}^{(i)}$ denote the cyclic classes of G containing elements of order p^i . Now from lower hook lemma 1 it follows that, for any $2 \leq i \leq e$ the cyclic class $\mathcal{C}_{m_i}^{(i)}$ is non-adjacent to one of the cyclic classes $\mathcal{C}_1^{(i-1)}, \mathcal{C}_2^{(i-1)}, \dots, \mathcal{C}_{m_{i-1}}^{(i-1)}$. By relabelling if necessary, we may assume that $\mathcal{C}_{m_i}^{(i)}$ is non-adjacent to $\mathcal{C}_1^{(i-1)}$.

Now for each $1 \leq i \leq e$, we set $\mathfrak{M}_i := (\mathcal{C}_1^{(i)}, \mathcal{C}_2^{(i)}, \dots, \mathcal{C}_{m_i}^{(i)})$, and from Definition 4 we construct the Hamiltonian path $\mathcal{H}(\mathfrak{M}_i)$ in $(\Gamma_G)^c$ on the vertex set $\Lambda(p^i; G)$. Then

for every $e \geq i \geq 2$, the vertices $\partial_1(\mathcal{H}(\mathfrak{M}_i))$ and $\partial_0(\mathcal{H}(\mathfrak{M}_{i-1}))$ are not adjacent in the power graph Γ_G , and consequently, they are connected by an edge β_i in the complement graph $(\Gamma_G)^c$. We can now construct the Hamiltonian path $\mathfrak{H}$ in $(\Gamma_G \setminus \{1_G\})^c$ defined by

$$\mathfrak{H} := (\mathcal{H}(\mathfrak{M}_e), \beta_e, \mathcal{H}(\mathfrak{M}_{e-1}), \beta_{e-1}, \mathcal{H}(\mathfrak{M}_{e-2}), \dots, \beta_2, \mathcal{H}(\mathfrak{M}_1))$$

□

Corollary 2 *Let G be a finite p -group. Then, $\lambda(\Gamma_G) = |G|$ if and only if G is neither cyclic, nor a generalized quaternion 2-group.*

Consequently, for a finite p -group G of exponent p^e we have

$$\lambda(\Gamma_G) = \begin{cases} 2(p^e - 1) & \text{if } G \text{ is cyclic} \\ |G| + 1 & \text{if } p = 2, G \text{ is generalized quaternion} \\ |G| & \text{otherwise} \end{cases}$$

Proof We will first show that for the semi-dihedral 2-group $G = \mathbb{SD}_{2^{e+1}}$ of exponent 2^e ($e \geq 3$) we have $\lambda(\Gamma_G) = |G|$ by constructing a Hamiltonian path in $(\Gamma_G \setminus \{1_G\})^c$. We are going to use the notations as given in the presentation of G in Lemma 1.

The cyclic maximal subgroup $\langle x \rangle \leq G$ has order 2^e and the elements outside this subgroup are given by $x^k y$ ($0 \leq k \leq 2^e - 1$). As noticed earlier, the order of $x^k y$ is 2 (resp. 4) if k is even (resp. k is odd).

We first consider the cyclic classes of elements of order 2 and 4 from the cyclic subgroup $\langle x \rangle$, namely $C_1^{(1)} = \{x^{2^{e-1}}\} \subseteq \Lambda(2; G)$ and $C_1^{(2)} = \{x^{2^{e-2}}, x^{3 \cdot 2^{e-2}}\} \subseteq \Lambda(4; G)$. Now we consider the sequence

$$T_1 := \left(y, x^{2^{e-1}}, x^2 y, x^{2^{e-2}}, x^4 y, x^{3 \cdot 2^{e-2}} \right).$$

Clearly this forms a path in the complement graph $(\Gamma_G)^c$. Now we fix an arbitrary labelling of the set $(G \setminus \langle x \rangle) \setminus \{y, x^2 y, x^4 y\}$ as follows

$$\mathcal{M} = \{x^{k_i} y : i = 1, 2, \dots, N\}$$

where $N = 2^e - 3$. Notice that the number of elements in $\mathcal{C} = \langle x \rangle \setminus \{1_G, x^{2^{e-1}}, x^{2^{e-2}}, x^{2^{e-1}+2^{e-2}}\}$ is $2^e - 4 = N - 1$.

We claim that for any pair of elements $a \in \mathcal{C}$ and $b \in \mathcal{M}$, the vertices a and b are not adjacent in Γ_G (and hence they are adjacent in the complement $(\Gamma_G)^c$). To see this, notice that a is a power of x and has order ≥ 8 , whereas b has order either 2 or 4. So if they are adjacent, then $b = a^m$ for some positive integer m , and consequently, $b \in \langle x \rangle$, a contradiction.

Now setting up an arbitrary enumeration of the elements of $\mathcal{C} = \{w_1, w_2, \dots, w_{N-1}\}$, we have the following path in $(\Gamma_G)^c$

$$T_2 = (x^{k_1} y, w_1, x^{k_2} y, w_2, \dots, x^{k_{N-1}} y, w_{N-1}, x^{k_N} y).$$

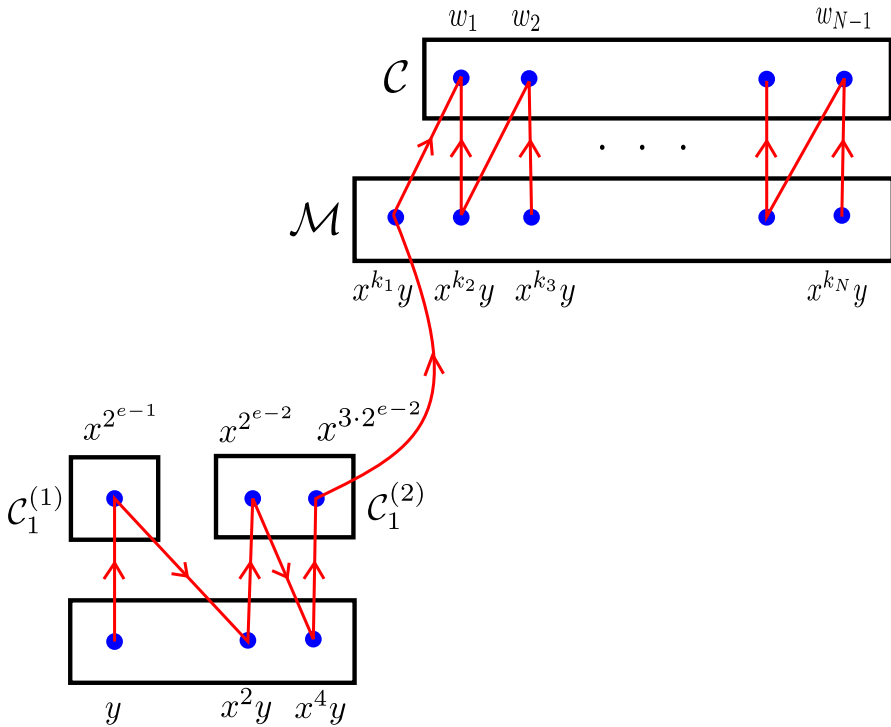


Fig. 3

Now a similar argument as above shows that the elements $x^{k_1}y$ and $x^{3 \cdot 2^{e-2}}$ are not adjacent in Γ_G . Hence, there exists an edge β in $(\Gamma_G)^c$ joining these two vertices. Consequently, the path (T_1, β, T_2) is a Hamiltonian path in $(\Gamma_G \setminus \{1_G\})^c$. This proves that $\lambda(\Gamma_G) = |G|$.

Now we are ready to prove the final statement. Let G be a finite p -group of exponent p^e . Assume that $\lambda(\Gamma_G) = |G|$. Since $\lambda(\Gamma_{C_{p^e}}) = 2(p^e - 1)$ (since $\Gamma_{C_{p^e}}$ is a complete graph) and $\lambda(\mathbb{SD}_{2^{e+1}}) = 2^{e+1} + 1$ (see [14, Example 3.3]), G cannot be these groups. Conversely, suppose G is neither cyclic, nor a generalized quaternion 2-group. If $m(p^i) \geq 2$ for every $1 \leq i \leq e$, then from Theorem 3, we are done. Hence, using Corollary 1, we have $p = 2$ and G is isomorphic to one of the groups $\mathbb{D}_{2^{e+1}}$ or $\mathbb{SD}_{2^{e+1}}$. If $G \cong \mathbb{D}_{2^{e+1}}$, then from [14, Example 3.2], the statement follows. Finally if $G \cong \mathbb{SD}_{2^{e+1}}$, then we have shown above that $\lambda(\Gamma_G) = |G|$.

If C_{p^e} is a finite cyclic group of order p^e , then $\Gamma_{C_{p^n}}$ is complete and consequently $\lambda(\Gamma_{C_{p^n}}) = 2(p^e - 1)$. Next, if G is the generalized quaternion group of order 2^n and exponent p^{n-1} , then $\lambda(\Gamma_G) = p^n + 1$ (see Example 3.3, [14]). This completes the proof. $\square$

Acknowledgements The authors would like to thank the anonymous referee for the improvement of the exposition of the article.

Data availability The article is self-contained, and there is no requirement of additional or supporting data for the readers.

References

1. Abawajy, J., Kelarev, A., Chowdhury, M.: Power graphs: a survey. *Electron. J. Graph Theory Appl. (EJGTA)* **1**(2), 125–147 (2013)
2. Berkovich, Y.: Groups of prime power order. Vol. 1, volume 46 of *De Gruyter Expositions in Mathematics*. Walter de Gruyter GmbH & Co. KG, Berlin (2008). (With a foreword by Zvonimir Janko)
3. Berkovič, J.G.: p -groups of finite order. *Sibirsk. Mat. Zh.* **9**, 1284–1306 (1968)
4. Cameron, P.J.: The power graph of a finite group, II. *J. Group Theory* **13**(6), 779–783 (2010)
5. Cameron, P.J., Ghosh, S.: The power graph of a finite group. *Discrete Math.* **311**(13), 1220–1222 (2011)
6. Chakrabarty, I., Ghosh, S., Sen, M.K.: Undirected power graphs of semigroups. *Semigroup Forum* **78**(3), 410–426 (2009)
7. Fernández-Alcober, G.A.: An introduction to finite p -groups: regular p -groups and groups of maximal class. In: *16th School of Algebra, Part I (Portuguese)* (Brasília, 2000), vol. 20, pp. 155–226 (2001)
8. Hale, W.K.: Frequency assignment: theory and application. *Proc. IEEE* **68**, 1497–1514 (1980)
9. Isaacs, I.M.: *Character theory of finite groups*. AMS Chelsea Publishing, Providence (2006). (Corrected reprint of the 1976 original [Academic Press, New York; MR0460423])
10. Kelarev, A.V., Quinn, S.J.: A combinatorial property and power graphs of groups. In: *Contributions to General Algebra*, vol. 12 (Vienna, 1999), pp. 229–235. Heyn, Klagenfurt (2000)
11. Kulakoff, A.: Über die Anzahl der eigentlichen Untergruppen und der Elemente von gegebener Ordnung in p -Gruppen. *Math. Ann.* **104**(1), 778–793 (1931)
12. Kumar, A., Selvaganesh, L., Cameron, P.J., Chelvam, T.T.: Recent developments on the power graph of finite groups—a survey. *AKCE Int. J. Graphs Comb.* **18**(2), 65–94 (2021)
13. Ma, X., Feng, M., Wang, K.: The strong metric dimension of the power graph of a finite group. *Discrete Appl. Math.* **239**, 159–164 (2018)
14. Ma, X., Feng, M., Wang, K.: Lambda number of the power graph of a finite group. *J. Algebraic Combin.* **53**(3), 743–754 (2021)
15. Ma, X., Walls, G.L., Wang, K.: Power graphs of (non)orientable genus two. *Comm. Algebra* **47**(1), 276–288 (2019)
16. Ma, X., Zhai, L.: Strong metric dimensions for power graphs of finite groups. *Comm. Algebra* **49**(11), 4577–4587 (2021)
17. Miller, G.A.: An extension of Sylow’s theorem. *Proc. Lond. Math. Soc.* **2**(2), 142–143 (1905)
18. Roberts, F.S.: T -colorings of graphs: recent results and open problems. *Discrete Math.* **93**(2–3), 229–245 (1991)
19. Yeh, R.K.: A survey on labeling graphs with a condition at distance two. *Discrete Math.* **306**(12), 1217–1231 (2006)

Publisher’s Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Terms and Conditions

Springer Nature journal content, brought to you courtesy of Springer Nature Customer Service Center GmbH (“Springer Nature”).

Springer Nature supports a reasonable amount of sharing of research papers by authors, subscribers and authorised users (“Users”), for small-scale personal, non-commercial use provided that all copyright, trade and service marks and other proprietary notices are maintained. By accessing, sharing, receiving or otherwise using the Springer Nature journal content you agree to these terms of use (“Terms”). For these purposes, Springer Nature considers academic use (by researchers and students) to be non-commercial.

These Terms are supplementary and will apply in addition to any applicable website terms and conditions, a relevant site licence or a personal subscription. These Terms will prevail over any conflict or ambiguity with regards to the relevant terms, a site licence or a personal subscription (to the extent of the conflict or ambiguity only). For Creative Commons-licensed articles, the terms of the Creative Commons license used will apply.

We collect and use personal data to provide access to the Springer Nature journal content. We may also use these personal data internally within ResearchGate and Springer Nature and as agreed share it, in an anonymised way, for purposes of tracking, analysis and reporting. We will not otherwise disclose your personal data outside the ResearchGate or the Springer Nature group of companies unless we have your permission as detailed in the Privacy Policy.

While Users may use the Springer Nature journal content for small scale, personal non-commercial use, it is important to note that Users may not:

1. use such content for the purpose of providing other users with access on a regular or large scale basis or as a means to circumvent access control;
2. use such content where to do so would be considered a criminal or statutory offence in any jurisdiction, or gives rise to civil liability, or is otherwise unlawful;
3. falsely or misleadingly imply or suggest endorsement, approval, sponsorship, or association unless explicitly agreed to by Springer Nature in writing;
4. use bots or other automated methods to access the content or redirect messages
5. override any security feature or exclusionary protocol; or
6. share the content in order to create substitute for Springer Nature products or services or a systematic database of Springer Nature journal content.

In line with the restriction against commercial use, Springer Nature does not permit the creation of a product or service that creates revenue, royalties, rent or income from our content or its inclusion as part of a paid for service or for other commercial gain. Springer Nature journal content cannot be used for inter-library loans and librarians may not upload Springer Nature journal content on a large scale into their, or any other, institutional repository.

These terms of use are reviewed regularly and may be amended at any time. Springer Nature is not obligated to publish any information or content on this website and may remove it or features or functionality at our sole discretion, at any time with or without notice. Springer Nature may revoke this licence to you at any time and remove access to any copies of the Springer Nature journal content which have been saved.

To the fullest extent permitted by law, Springer Nature makes no warranties, representations or guarantees to Users, either express or implied with respect to the Springer nature journal content and all parties disclaim and waive any implied warranties or warranties imposed by law, including merchantability or fitness for any particular purpose.

Please note that these rights do not automatically extend to content, data or other material published by Springer Nature that may be licensed from third parties.

If you would like to use or distribute our Springer Nature journal content to a wider audience or on a regular basis or in any other manner not expressly permitted by these Terms, please contact Springer Nature at

onlineservice@springernature.com